

From Orbital Debris Risk to Policy Action: Economic Decision Support for Space Sustainability

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Abstract

The expansion of satellite activity in low Earth orbit has made space sustainability a coupled engineering, economic, and policy problem. While debris-environment models can estimate collision risk, object growth, and orbital capacity degradation, these outputs are often not expressed in a form that directly supports regulatory design, operator incentives, or long-term governance decisions. This research addresses the gap between technical orbital debris simulation and policy-relevant decision-making by developing Catalyst, an economics-informed decision-support framework for evaluating how mission and policy choices affect the shared orbital environment. Catalyst uses the Embry-Riddle MIT Orbital Capacity Assessment Toolbox Monte Carlo (EMOCAT-MC) framework as its technical simulation engine and extends it into a structured policy-analysis workflow. The framework ingests local satellite catalog and TLE data, executes reproducible seeded Monte Carlo scenarios, compares orbital-environment outcomes across controlled assumptions, and maps simulation outputs to sustainability metrics associated with collision risk, shell-level density growth, fragmentation footprint, and multi-shell critical density. The central research question is how technical measures of orbital crowding and fragmentation risk can be converted into interpretable policy and economic indicators for regulators, satellite operators, and mission planners. The methodology evaluates representative low Earth orbit scenarios under varying assumptions about launch activity, post-mission disposal compliance, fragmentation likelihood, and mitigation behavior. For each scenario, Catalyst compares object population growth, collision-risk evolution, shell-level density behavior, and fragmentation consequences. These outputs are then interpreted through a policy/economic layer that estimates the relative burden imposed by different operating assumptions and identifies which mitigation choices produce the largest reduction in long-term environmental risk. This approach treats orbital capacity as a shared resource and frames debris generation as an externality that can be quantified through simulation-informed decision metrics. Work completed to date includes the design of the Catalyst mission-analysis architecture, implementation of a backend workflow for local-truth catalog ingestion, reproducible MOCAT-based scenario execution, preliminary run comparison, and development of a policy-summary layer that connects simulation results to sustainability conclusions. The next phase will validate the framework using controlled LEO case studies comparing baseline growth, improved disposal compliance, reduced fragmentation probability, and enhanced mitigation behavior. Expected results will identify which policy levers most strongly affect long-term orbital capacity and demonstrate how debris-environment models can support more transparent space sustainability decisions. The contribution of this work is a bridge between astrodynamics-based debris modeling and economics-informed space governance. By translating Monte Carlo orbital capacity analysis into decision-support metrics, Catalyst provides a framework for evaluating not only what may happen in the orbital environment, but also how specific policy and operator choices contribute to or reduce long-term risk. This aligns with the need for analytical tools that help decision-makers manage orbital capacity as a finite, shared resource.

1. Introduction

Low Earth orbit is an engineering environment, an input to the space economy, and a shared resource whose quality is affected by cumulative use. Collision hazards arise from the interaction of active spacecraft, derelict objects, and fragments, while the resulting tracking, avoidance, collision, and remediation costs are distributed across operators, governments, and future users [1]. Debris therefore has the structure of a negative externality: a mission can capture the private value of deployment while transferring part of its expected tracking, avoidance, collision, and remediation burden to others [5-8].

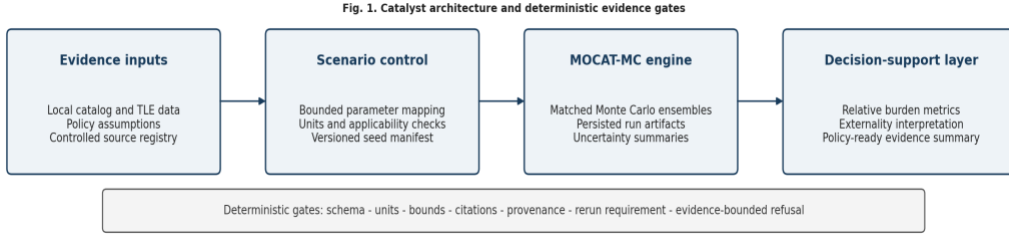
This externality is difficult to govern because simulation evidence and policy decisions use different analytical languages. Long-term debris models return ensembles of population trajectories, collision counts, shell densities, and uncertainty distributions. Licensing, investment, and remediation decisions require comparisons among concrete choices, such as changes in post-mission disposal (PMD) reliability, passivation, launch cadence, and active debris removal. A credible decision-support system must preserve the physics and uncertainty of the simulation while making the connection between an operating assumption, a measurable environmental consequence, and a policy instrument explicit. Catalyst addresses this translation problem by coupling a reproducible MOCAT-MC analysis workflow with an evidence-constrained Code World Model (CWM) assistant. MOCAT-MC provides the stochastic environment model [2], while Catalyst adds versioned scenario construction, persisted run artifacts, bounded mappings between governance concepts and backend parameters, and policy/economic interpretation. The contribution is not an automated regulator or a monetary valuation model. It is an auditable analytical bridge that converts matched simulation comparisons into relative burden indicators while refusing conclusions that exceed the available evidence.

2. Catalyst Decision-Support Framework

2.1 Reproducible evidence-to-action workflow

The workflow begins with local-truth catalog and two-line element data rather than an unconstrained narrative description of the orbital environment. Each case is represented by a versioned scenario manifest containing the input population, simulation duration, parameter values, seed set, software revision, and evidence sources. Policy concepts are then resolved through a bounded registry. For example, post-mission disposal (PMD) compliance maps to the backend parameter PMD success rate, launch restraint maps to launch rate multiplier, and fragmentation passivation maps to the MOCAT-MC fragmentation probability parameter. The registry enforces type, unit, minimum, maximum, and transformation rules and marks each substantive adjustment as requiring a new simulation run.

Controlled scenario pairs are evaluated with matched seed sets so that changes in the output distribution can be attributed to the policy assumption rather than to arbitrary differences in stochastic realization. The backend persists the resulting population histories, collision and conjunction indicators, shell-level density measures, fragmentation consequences, run metadata, and uncertainty summaries. Hashes and fixed manifests connect every conclusion to the exact evidence packet from which it was derived. This design makes reproducibility part of the decision product rather than a separate documentation exercise.



2.2 Economic translation without false precision

The economic layer treats orbital capacity as a common-use environmental asset whose governance requires explicit rules, monitoring, and accountability [15]. Classic debris economics shows why mitigation can be underprovided when operators do not bear the full expected cost of the risk they create [5,6]. Dynamic models further show that private deployment incentives can diverge from socially efficient use [6], and orbital-use fees have been proposed as one mechanism for internalizing future collision risk [7]. The OECD similarly emphasizes the need for economic evidence that links debris exposure to public action [8]. Catalyst operationalizes this literature at the scenario level by retaining a vector of physical changes rather than collapsing them immediately into a single price.

$$\Delta z_j = [\Delta N_j, \Delta R_j, \Delta D_j, \Delta F_j, \Delta K_j]$$

For policy alternative j , ΔN denotes the change in long-term object population, ΔR the change in collision or conjunction risk, ΔD the change in shell-level density, ΔF the change in fragmentation burden, and ΔK the change in proximity to multi-shell criticality [3,4]. Each component is normalized against a matched baseline and accompanied by ensemble dispersion. The resulting vector can support dominance tests, threshold screening, or a decision-maker-supplied weighting function. Catalyst does not assign normative weights silently. A regulator may prioritize preservation of a persistent shell, an operator may emphasize maneuver and replacement burden, and a public insurer may emphasize expected loss. Keeping those preferences explicit prevents a model-generated score from being mistaken for an empirically validated social welfare function.

The policy summary therefore distinguishes three claims: what changed in the simulated environment, who would plausibly bear the resulting burden, and which intervention is institutionally capable of affecting the mapped parameter. This separation is central to transparent economic decision support. It enables comparisons among mitigation choices without implying that simulation output alone establishes a legally binding standard, an insurance premium, or an optimal fee.

2.3 Controlled policy cases and physical campaign

The burden vector is constructed from matched scenario differences rather than absolute values taken from unrelated runs. For metric m , policy case j , seed s , and time t , Catalyst computes $\Delta x_{jms}(t) = x_{jms}(t) - x_{0ms}(t)$, where case 0 is the registered baseline. It reports the ensemble median, dispersion, and the fraction of seeds for which the sign of the effect is preserved. A claimed improvement is therefore supported only when the direction is stable across a pre-specified portion of the ensemble, and the magnitude exceeds a decision-relevant tolerance. This prevents a small stochastic fluctuation from being presented as a policy effect.

$$B_j(w) = S_j = \sum_{m=1}^M w_m q_m(\Delta z_{jm}), w_m \geq 0, \quad \sum_{m=1}^M w_m = 1$$

A scalar burden score may be calculated only when a decision-maker supplies the weights w and the normalization functions q . The score is not a universal social-welfare measure. It is a transparent sensitivity device that shows how a ranking changes under alternative priorities. Catalyst also returns the unweighted vector and a Pareto set. If one case lowers collision exposure but raises launch-access or implementation burden, it remains non-dominated rather than being forced into a single ordinal ranking. Threshold analysis is handled separately: a regulator can ask whether a case remains below a shell-density, fragmentation-footprint, or criticality limit with a specified confidence level, while an operator can ask which feasible mission design produces the greatest marginal reduction per unit of mitigation effort.

Table 1. Decision metrics retained by the Catalyst policy/economic layer.

Term	Physical quantity	Economic burden represented	Decision use
ΔN	Long-term population change, separated by active, derelict, and fragment classes.	Future tracking, avoidance, replacement, and remediation exposure.	Population-growth screening; contribution accounting.
ΔR	Change in collision or conjunction-risk indicators and their time evolution.	Expected operational disruption and loss exposure shared across operators.	Risk tiers; assurance, insurance, and maneuver requirements.
ΔD	Shell-level density and occupancy growth relative to the matched baseline.	Congestion and reduced option value for future users of a shell.	Shell-specific licensing thresholds and capacity monitoring.

Term	Physical quantity	Economic burden represented	Decision use
ΔF	Fragmentation footprint, persistence, and cross-shell propagation consequences.	Long-lived external costs that may extend beyond the initiating operator and shell.	Passivation standards, breakup prevention, and remediation priority.
ΔK	Distance to or exceedance of multi-shell critical-density conditions.	Systemic loss of orbital resilience and correlated risk across shells.	Precautionary screening and adaptive governance triggers.

Metrics remain disaggregated so that physical consequences, burden incidence, and policy preferences are not conflated.

The four policy cases are formulated as interventions against one common baseline. Each alternative changes only registered assumptions, preserves the initial population and seed schedule, and returns both central tendency and dispersion. Catalyst first tests for robust dominance: alternative j is preferred to k only when its burden vector is no worse on every decision-relevant component and is materially better on at least one component after uncertainty is considered. When one alternative reduces collision risk but increases lifecycle or access burden, the system reports a non-dominated frontier rather than manufacturing a single ranking. This rule is particularly important because launch restraint, disposal compliance, and fragmentation prevention act through different mechanisms and on different time scales.

Table 2. Controlled policy cases and required evidence.

Policy case	Registered control	Economic mechanism	Required comparison
Launch activity	launch_rate_multiplier	Changes new occupancy and future collision opportunities; does not remove legacy stock.	Matched baseline; population, density, risk, and criticality over time.
Disposal compliance	pmd_success_rate	Reduces the stock of failed objects and transfers some prevention cost to the operator.	Active/inactive populations, collision risk, footprint, and uncertainty.
Fragmentation control	P_{frag} through registered probability or passivation controls	Reduces expected fragment creation and persistent cross-shell burden.	Breakup frequency, fragment burden, shell density, and criticality.
Enhanced mitigation	Bounded bundle of registered controls; no invented parameter	Combines private operational effort to reduce shared environmental burden.	Pareto comparison against single-lever cases and matched baseline.

The physical campaign, not the language-model benchmark, must establish the direction and magnitude of these intervention effects.

The physical campaign is organized as a matched, staged experiment. Stage one establishes the baseline distribution and confirms numerical stability across the seed set. Stage two changes one registered policy lever at a time so that the marginal effect of launch activity, disposal compliance, and fragmentation control can be identified. Stage three evaluates the enhanced-mitigation bundle and selected two-factor combinations to test complementarity or substitution among interventions. All cases retain the same initial population, propagation configuration, output definitions, and reporting horizon. A result is not attributed to a policy lever unless the relevant run manifest differs from the baseline only in the registered transformation and any explicitly documented dependencies.

The experiment is designed to test four expectations rather than assume them. Launch restraint should affect the inflow of new objects and near-term occupancy but cannot directly remove inherited debris. Improved disposal compliance should reduce the future derelict population, with environmental benefits that may emerge after a lag and vary by shell persistence. Fragmentation prevention should primarily affect the upper tail of population and risk outcomes because rare breakup events can generate persistent cross-shell consequences. The enhanced-mitigation case tests whether these mechanisms reinforce one another or whether one lever supplies most of the attainable reduction. Catalyst will report when the ensemble does not distinguish the cases; absence of a robust difference is a decision result, not a failure to produce a ranking. For each case, the policy summary will identify the time at which effects first become distinguishable, the shells in which the effect is concentrated, the stability of the sign across seeds, and the sensitivity of the conclusion to uncertain inputs. This temporal and spatial decomposition is necessary for policy design. A measure that reduces long-horizon criticality but has little near-term effect may support a structural licensing rule, while a measure that rapidly reduces conjunction burden may support an operational requirement. The planned campaign thus evaluates not only whether a lever improves the environment, but also the mechanism, timing, and institutional relevance of the improvement.

3. Evaluation Design

Catalyst was post-trained from the public CWM-SFT checkpoint, an open-weights Code World Model designed for reasoning and planning in executable computational environments [9]. The final evaluation examined whether domain adaptation improved grounded orbital-debris policy reasoning, not whether the underlying orbital physics was correct. Two complementary cohorts were retained separately because their metrics answer different questions. The first comprised 1,252 unique held out and regression cases across 20 task types. The second was a paired 800-case evidence-stress campaign containing eight 100-case sections spanning backend interpretation, governance, economics, orbital risk, MOCAT and Monte Carlo reasoning, tool execution, provenance safety, and end-to-end decision support [14].

For the paired campaign, Base and Catalyst received identical ordered prompts, model-visible evidence, evidence hashes, and greedy decoding settings. The 800 cases were divided into eight equal research sections and included supported-answer, partial-

evidence, and insufficient-evidence tracks. Four axes were scored independently: defensibility, full task success, usable partial success, and operational contract completion. Defensibility includes a correct answer, a supported partial answer, or an explicit evidence-bounded refusal. Full task success requires complete semantic performance. Operational contract completion additionally requires complete generation, valid schema, citations, and required interface elements. The distinction prevents safe refusal from being mislabeled as complete accuracy.

The older cohort tests whether post-training learned the repository-specific mappings and reasoning conventions required by the architecture. It includes scenario construction, tool planning, backend result interpretation, code-path and parameter attribution, threshold comparison, economic reasoning, policy application, uncertainty analysis, and adversarial refusal. The paired cohort instead stress-tests behavior when the evidence packet is incomplete, conflicting, or insufficient. Because the scoring definitions differ, the cohorts are reported separately rather than pooled into a synthetic accuracy value. This separation is essential to the paper's claim: Catalyst must both answer supported questions and stop at the evidentiary boundary when the physical or legal record does not justify a conclusion. The 800 Catalyst outcomes decompose into 543 full successes, 73 qualified or partial successes, 169 safe non-answers or over-refusals, and 15 defensibility misses. This decomposition is more informative than a single headline rate. It shows that post-training sharply reduced materially unsupported answers while increasing complete task performance, but it also made the model more conservative. The intended production architecture therefore combines the learned reasoning layer with deterministic schema, unit, bound, citation, provenance, and execution gates rather than treating the model output as self-validating.

4. Results

Post-training produced a large improvement in grounded domain performance. On the older cohort, the evaluator composite increased from 40.6% to 96.3%, and production-verifier acceptance on 1,001 direct grounded cases increased from 1.1% to 95.8%. The report's deterministic functional re-adjudication found correct performance on 24 of 24 economic reasoning cases, 49 of 49 policy application cases, 426 of 451 backend interpretation cases, and 48 of 53 uncertainty and Monte Carlo cases. These results demonstrate that Catalyst learned the target mappings and evidence conventions, although the functional re-adjudication is not an independently blinded human-expert estimate.

Capabilities that directly support the abstract were also evaluated separately. Code-path and parameter attribution was functionally correct in 64 of 64 cases; scenario clarification and construction were correct in 14 of 14 and 13 of 13 cases; tool planning was correct in 81 of 81 cases; and full orchestration was correct in 12 of 13 cases. Governance adjustment was functionally correct in 58 of 68 cases, while safe full mapping was achieved in 25 of 34 mapping cases. These results support the feasibility of converting a policy concept into a bounded backend adjustment and then interpreting a persisted run, but the mapping and execution errors justify retaining deterministic bounds checks and requiring a new simulation whenever a substantive parameter changes. The paired stress campaign provides the stronger safety comparison. Defensible or safely bounded responses increased from 82.5% to 98.1%, a 15.6 percentage-point gain. There were 133 Base-fail/Catalyst-pass pairs and eight Base-pass/Catalyst-fail pairs; the exact McNemar p-value was 2.41×10^{-30} , and Catalyst's Wilson 95% interval was 96.9% to 98.9%. Full task success also increased, from 56.1% to 67.9%, confirming that the gain was not solely produced by refusal behavior.

The improvements were concentrated in the domains most relevant to the abstract. Full task success increased from 67% to 81% for economics, incentives, and decision cases; from 49% to 85% for orbital and collision cases; from 47% to 74% for evidence and provenance cases; and from 55% to 70% for cross-domain end-to-end cases. Catalyst reached 99% defensibility in the economics, orbital, MOCAT, and policy sections. This supports the claim that the framework can translate controlled backend evidence into bounded economic and governance explanations.

5. From Simulation Evidence to Policy Action

Catalyst's policy value lies in comparative attribution. A baseline and an intervention are evaluated with the same initial population, time horizon, seed set, and output definitions. The system can then state that a change in disposal compliance, fragmentation likelihood, or launch rate produced a measured shift in the burden vector, while preserving the uncertainty and the exact parameter transformation. The decision product contains five linked elements: the registered intervention, the physical effect and its ensemble uncertainty, the affected orbital shells and time horizon, the parties likely to bear the burden, and the institutional instrument capable of changing the input. This structure prevents a technically correct simulation result from being converted into an unsupported policy recommendation.

Burden incidence is treated explicitly. Operators may bear design, maneuver, disposal, or insurance costs; other operators may bear additional conjunction-screening and collision-avoidance costs; governments may bear surveillance, remediation, and liability exposure; and future entrants may bear reduced access or higher mission-design costs. Catalyst does not determine that distribution from object counts alone. It records the causal pathway supported by the scenario and labels transfers or normative judgments that require external economic or legal evidence. This distinction is particularly important when comparing prevention and remediation: the actor paying for mitigation may not be the actor receiving most of the long-term benefit.

This architecture enables four decision uses. First, licensing can become more performance sensitive. Existing mitigation frameworks emphasize limiting debris release, avoiding breakups, and disposing of spacecraft after mission [10-13]. Catalyst can connect a proposed compliance level or passivation claim to the backend parameter that represents it, test the claim under a defined traffic scenario, and report the incremental burden relative to a baseline. The output does not replace legal judgment, but it gives regulators a reproducible basis for differentiating missions whose expected long-term effects are materially different. Second, financial assurance and insurance can be informed by environmental contribution rather than mission value alone. A high-value spacecraft may be well controlled and rapidly disposed, while a less expensive but massive, non-maneuverable object in a

persistent shell may impose a larger expected external burden. Scenario-based indicators can support risk tiers, bond requirements, or premium incentives for reliable disposal, maneuverability, data sharing, and passivation. Actual prices would still require loss data, causal calibration, and an institutional choice about which external costs should be internalized.

Third, remediation priorities can be evaluated by marginal environmental benefit. Candidate removal actions can be modeled as controlled changes to the initial population and compared through the same seed registry. The relevant decision quantity is not simply the number of objects removed, but the reduction in collision exposure, fragmentation footprint, and cross-shell criticality per unit of remediation cost. This directly addresses the free-rider problem identified in debris economics: the party financing removal may not capture all the benefit generated for other users [5,6]. Fourth, adaptive governance can be built around repeated evidence updates. Catalog quality, constellation plans, maneuver practices, and atmospheric conditions change. A policy threshold derived from one snapshot should therefore be re-evaluated as the evidence packet changes. Catalyst's persisted manifests and hashes allow a regulator or operator to identify which inputs changed, rerun matched ensembles, and distinguish a real environmental shift from a change in modeling assumptions. This is especially important for policies such as the FCC's five-year disposal rule, which translate a general sustainability objective into an operational requirement [11].

6. Limitations and Validation Requirements

The planned physical validation will pre-register the baseline population, launch schedule, time horizon, seed set, parameter transformations, outcome definitions, and convergence rule. Each intervention will be run against the same seed registry, and results will be reported as trajectories and distributions rather than single terminal values. Sensitivity tests will vary catalog uncertainty, launch assumptions, disposal reliability, and fragmentation parameters within defensible ranges. A policy lever will be described as effective only when its environmental advantage is robust to those tests and when the claimed effect can be traced to the registered parameter change rather than to a different initial population or software state.

The evaluation establishes model capability, not physical truth. It does not validate MOCAT-MC calibration, establish a stabilization threshold, prove a nonlinear fragmentation response, or determine that launch restraint is insufficient. Those conclusions require actual matched, multi-seed simulation campaigns with convergence checks, ensemble dispersion, and preserved artifacts. The present evidence therefore supports the architecture and its reasoning layer, while the abstract's planned LEO case studies remain the necessary test of environmental and policy-effect claims.

The assistant also remains imperfect. The 98.1% defensibility result includes 169 safe non-answers or over-refusals; full task success was 67.9%, and complete operational contract success was 69.5%. Tool and scenario execution was the only section in which full success declined, from 67% to 55%, and Catalyst produced 17 incomplete or truncated outputs compared with three for Base. The 800-case campaign was adjudicated through an auditable single-reviewer AI process rather than an independently blinded legal or policy panel. Production use therefore requires deterministic validators, execution checks, and human review. Catalyst should be understood as constrained decision support, not autonomous policy authority.

7. Discussion

The central methodological advance is the preservation of an evidentiary chain from mission assumption to policy statement. Debris models are often treated as technical forecasting tools, while economic instruments are discussed at a level detached from model parameters. Catalyst links the two without erasing the distinction between positive and normative analysis. The simulation estimates consequences under stated assumptions; the economic layer identifies private and external burdens; the policy layer identifies feasible instruments and records the values chosen by the decision-maker. This separation also makes disagreement productive. Two regulators can apply different risk tolerances or distributional weights to the same physical result while still sharing a common factual record. An operator can challenge an input population, a fragment-yield assumption, or a policy weight without contesting every other element of the analysis. In this sense, Catalyst is not only a translation interface but an institutional design for transparent contestability. The framework's novelty is therefore not the claim that a language model can replace orbital analysis or economic valuation. Its novelty is the construction of a controlled interface between them: policy language is mapped to executable parameters; stochastic outputs remain linked to their run artifacts; economic interpretation is expressed as relative burden rather than fabricated price; and unsupported conclusions trigger an evidence-bounded response. This combination is necessary for decision support in a domain where both physical uncertainty and institutional disagreement are unavoidable.

8. Conclusion

Space sustainability decisions require more than accurate orbital simulation. They require a defensible account of how a mission choice changes the shared environment, how the resulting burden is distributed, and which policy lever can address it. Catalyst provides that account through a reproducible chain of local-truth data ingestion, bounded scenario construction, seeded MOCAT-MC comparison, persisted evidence, and economics-informed interpretation. The completed model evaluation supports the framework's decision-support claim. Catalyst substantially improved grounded backend interpretation, economic reasoning, orbital-risk reasoning, provenance discipline, and end-to-end policy analysis over the Base CWM-SFT model. At the same time, the separation between defensibility, completeness, and operational contract performance exposes the remaining reliability gap rather than concealing it. The next empirical requirement is a controlled physical campaign comparing baseline traffic with improved disposal compliance, reduced fragmentation probability, and enhanced mitigation behavior. Only those simulations can establish which intervention produces the largest long-term reduction in orbital burden. When coupled to that evidence, Catalyst can help regulators, operators, and mission planners move from generalized concern about debris to transparent, reproducible, and economically coherent policy action.

9. References

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